WORKED SOLUTION

STEP I 1992, Question 1

Pure • Calendar Arithmetic Modulo 7

> **STEP I 1992, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/92-s1-q1

The key idea

Days of the week repeat every 7 days, so all we need is the number of days between the two dates, reduced modulo 7. The trick that makes this painless is that $365 = 52 \times 7 + 1$: an ordinary year shifts the day of the week by one, and a leap year by two. So we count whole years and leap days first, then deal with the odd months at the end. "Carefully" means being precise about which February 29ths actually fall inside the interval, and about the one century year that might trip us up.

Solution

We split the interval from 3 April 1905 to 26 June 1992 into two pieces:

$$3 \text{ April } 1905 \longrightarrow 3 \text{ April } 1992 \longrightarrow 26 \text{ June } 1992.$$

Whole years: 3 April 1905 to 3 April 1992

This is $1992 - 1905 = 87$ whole years. Each such year (from 3 April to the next 3 April) contains 365 days, plus one extra day if a 29 February falls inside it.

Which 29 Februaries lie between 3 April 1905 and 3 April 1992? The leap years in this range are those divisible by 4, namely

$$1908,\ 1912,\ \ldots,\ 1988,\ 1992.$$

The 29 February of 1904 came before 3 April 1905, so it is not counted; the 29 February of 1992 comes before 3 April 1992, so it is counted. That gives

$$\frac{1992 - 1908}{4} + 1 = 21 + 1 = 22$$

leap days.

Ask yourself Is every year divisible by 4 in this range really a leap year?

In the Gregorian calendar, century years are leap years only if they are divisible by 400, so 1900 was *not* a leap year (and 2000 was). But neither of these lies in our interval, so the simple "divisible by 4" rule (the one in the rhyme) is exactly right here. Had the question gone back before 1900, we would have had to drop 29 February 1900.

So the number of days from 3 April 1905 to 3 April 1992 is

$$87 \times 365 + 22 = 31\,755 + 22 = 31\,777.$$

Working modulo 7 is quicker: since $365 \equiv 1 \pmod 7$, this is

$$87 + 22 = 109 = 15 \times 7 + 4 \equiv 4 \pmod 7.$$

The odd months: 3 April 1992 to 26 June 1992

Using the rhyme, April has 30 days and May has 31, so

$$\begin{aligned} 3\text{ April} &\to 3\text{ May} : 30\text{ days},\\ 3\text{ May} &\to 3\text{ June} : 31\text{ days},\\ 3\text{ June} &\to 26\text{ June} : 23\text{ days},\end{aligned}$$

a total of $30 + 31 + 23 = 84 = 12 \times 7$ days, which is $\equiv 0 \pmod 7$. Pleasingly, 3 April 1992 and 26 June 1992 fall on the same day of the week.

Putting it together

The total gap is

$$31\,777 + 84 = 31\,861 = 4551 \times 7 + 4 \text{ days},$$

so 3 April 1905 is 4 days of the week *earlier* than 26 June 1992. Counting back four days from Friday: Thursday, Wednesday, Tuesday, Monday. Hence

$$3\text{ April } 1905 \text{ was a Monday.}$$

Checks

- A computer counted the days between the two dates directly, month by month, and found 31 861 days, which is $4 \pmod 7$; a computer calendar also gives Monday for 3 April 1905 (and Friday for 26 June 1992).
- Sanity check on direction: going *forward* 4 days from Monday gives Friday, as it should.
- Check the leap-day count another way: 87 years contain roughly $87/4 \approx 21.75$ leap days, so 22 is about right.

Common slips

- Including 29 February 1904 (which is before the start date) or excluding 29 February 1992 (which is before the end date): each gives an answer one day out.
- Counting back instead of forward, or vice versa: the remainder 4 means 1905 is four days *before* Friday in the weekly cycle, not after.
- Treating 1900 as a leap year when extending the method to earlier dates.

Beyond STEP

This is arithmetic modulo 7: the key fact $365 \equiv 1 \pmod 7$ turns a large count into a small one. Congruences, and their use in calendar puzzles such as Zeller's congruence for the day of the week of any date, are a standard part of a first course in number theory. Davenport's *The Higher Arithmetic* is a lovely introduction to congruences and much more.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.